1. Clean and explore the data, discover natural groupings of cars, then build a classifier and check whether the clusters agree with a known label (am: automatic vs. manual transmission).

Dataset: mtcars (built-in)

Tasks:

1. **Preprocess:** Check for missing values, then standardize all numeric columns (scale()).
2. **Explore:** Build a correlation matrix of the numeric variables and identify the two most correlated with mpg.
3. **Cluster:** Run k-means with k = 2 on the scaled data. Compare the cluster labels against the actual am column using a contingency table — do the clusters roughly correspond to transmission type?
4. **Classify:** Split the data 70/30, train a decision tree (rpart) to predict am from the other variables, and evaluate it with a confusion matrix.
5. **Interpret:** In 2–3 lines, compare what the *unsupervised* clustering found vs. what the *supervised* classifier learned — do they agree on which variables matter?

2. Convert a built-in contingency table into transactional and tabular form, mine association rules about passenger survival, then build a classifier to predict survival.

Dataset: Titanic (built-in 4-way table: Class, Sex, Age, Survived)

Tasks:

1. **Transform:** Convert the Titanic table into a data frame of individual passenger records using as.data.frame() and expanding counts (Freq) into repeated rows.
2. **Association rules:** Convert the data frame into transactions (arules::transactions) and run apriori() to find rules whose right-hand side is Survived=Yes. Sort by confidence and inspect the top 5.
3. **Classify:** Train a Naive Bayes model (e1071::naiveBayes) to predict Survived from Class, Sex, and Age. Split 70/30 and evaluate with a confusion matrix.
4. **Compare:** Do the strongest association rules (e.g. which class/sex combinations survive most) match what the Naive Bayes model considers most important?
5. **Interpret:** In 2–3 lines, state which passenger profile had the highest survival likelihood, according to both methods.